ASHRAE GreenTip #6-4

Dynamic Glazing

GENERAL DESCRIPTION

Dynamic architectural glazing uses electrochromic technology to darken the interior of a piece of glass, enabling a thin coating to change from clear to tinted when small electrical voltage is applied. A demonstration of this technology is eyeglasses that darken when one goes outdoors. Dynamic glass for architectural applications can transform into four darkness levels depending on the season, being darker in summer to keep solar gain out and minimize glare. In addition, because the glazing becomes darker in the summer when exposed to sun, its use will generally minimize radiation heat transfer and help improve thermal comfort in the space,. Typically, the tint corresponds to 4%, 20%, 40%, and 60% visible light transmittance T_{vis} depending on manufacturers. Dynamical glazing promotes a reduced carbon footprint, helps achieve LEED certifications, and reduces energy costs by reducing solar heat gain (i.e., the amount of heat that enters a glass structure naturally from the sun). The dynamic glazing is usually controlled by a low voltage current and can be controlled either manually through a switch or linked into an automatic building management system (BMS).

APPLICABILITY

Dynamic glazing is usually the most feasible on large glazing facades not do not face north. It works well on east and west orientations since these are difficult to horizontally shade. It could also be used efficiently on south facades that are not shaded.

PROS

- Reduced energy consumption
- Reduced glare
- Improved comfort
- Reduced carbon footprint

CONS

- Higher first cost
- Requires electrical voltage to operate

ASHRAE GreenTip #9-12

Desiccant Cooling and Dehumidification

GENERAL DESCRIPTION

There are two basic types of open-cycle desiccant process: solid and liquid desiccant. Each of these processes has several forms, and these should be investigated to determine the most appropriate for the particular application. Both of these systems need to have the air being conditioned come in good contact with the desiccant, during which moisture is absorbed from the air and the temperatures of both the air and desiccant are coincidently raised.

The moisture absorption process is caused by desiccant having a lower surface vapor pressure than the air. As the temperature of the desiccant rises, its vapor pressure rises, and its useful absorption capability lessens. Some systems, particularly liquid types, have cooling of air and desiccant coincident with dehumidification. This can allow the need for less space and equipment.

The dehumidified air then has to be cooled by other means. Two supply air arrangements are available. One method uses a mixture of recycled return air and dehumidified outdoor ventilation air as supply air to the building. Moisture and contaminants such as volatile organic compounds can be absorbed by the desiccant and recycled; particles of solid or liquid desiccant may also be carried over into the ducts and to building occupants.

The other arrangement combines energy recovery from building exhaust air that is typically much cooler and less humid than outdoor ventilation air. By dehumidifying the exhaust air to a sufficiently low humidity ratio (i.e., moisture content), it can be used to indirectly cool outdoor air for supply to the building that has not contacted desiccant. Using the recovered energy, this arrangement can be used to process the total ventilation air requirement, even up to 100% from outside.

So that desiccant can be reused, it has to be redried by a heating process generally called *reactivation* or *regeneration* (for solid types) or *reconcentration* (for liquid types). The redrying can be either direct (by contact with heated outdoor air with the desiccant) or indirect. Indirect may be preferable, particularly in high humid climates, because of the higher temperature needed to maximize the vapor difference and drying potential. The energy storage benefit for liquid desiccant was discussed in GreenTip #9-12.

Rotary desiccant dehumidifiers use solid desiccants such as silica gel to attract water vapor from the moist air. Humid air, generally referred to as *process air*, is dehumidified in one part of the desiccant bed while a different part of the bed is dried for reuse by a second airstream known as *reactivation air*. The desiccant rotates slowly between these two airstreams, so that dry, high-capacity desiccant leaving the reactivation air is available to remove moisture from the moist process air.

Process air that passes through the bed more slowly is dried more deeply, so for air requiring a lower dew point, a larger unit (and slower velocity) is required. The reactivation air inlet temperature changes the outlet moisture content of the process air. In turn, if the designer needs dry air, it is generally more economical to use high reactivation temperatures. On the other hand, if the leaving humidity need not be especially low, inexpensive, low-grade heat sources (e.g., waste heat or rejected cogeneration heat) can be used.

The process air outlet temperature is higher than the inlet temperature primarily because the heat of sorption of the moisture removed is converted to sensible heat. The outlet temperature rises roughly in proportion to the amount of moisture that is removed. In most comfort applications, provisions must be made to remove excess sensible heat from the process air following reactivation. Cooling is accomplished with cooling coils, and the source of this cooling affects the operating economics of the system.

WHEN/WHERE IT'S APPLICABLE

In general, applications that require a dew point at or below 40°F (4.3°C) may be candidates for active desiccant dehumidification. Examples of such candidates include facilities handling hygroscopic materials; film drying; the manufacturing of candy, chocolate, or chewing gum; the manufacturing of drugs and chemicals; the manufacturing of plastic materials; packaging of moisture-sensitive products; and the manufacturing of electronics. Supermarkets often use desiccant dehumidification to avoid condensation on refrigerated casework. And when there is a need for a lower dew point and a convenient source of low-grade heat for reactivation is available, rotary desiccant dehumidifiers can be especially economical.

glazed and unglazed units (Weiss et al. 2008). In North America (i.e., United States and Canada), swimming pool heating is dominant with an installed capacity of about 55 × 10^9 Btu/h (16 GW_{th}) of unglazed plastic collectors. In other countries, flat-plate and evacuated-tube collectors are used to generate hot water for sanitary use and space heating. Solar energy is also used for applications such as industrial or agricultural processes.

For many, a key impediment to increased solar use is economics. The cost of some solar technologies is perceived to be high compared to the fossil-based energy source it is offsetting. For example, while the simple payback of solar PV systems tends to be rather long (although much improvement has occurred over the past couple of decades), the recent increase in the cost of energy and advances in solar energy justify a fresh look at the applications and the engineering behind those applications. In addition, public policy in many areas encourages solar and other renewable energy applications through tax incentives and encouraging or requiring repurchase of excess electrical energy generated by the utility provider.

The applicability and, consequently, the economics and public policy incentives available with different solar energy system types and applications depend greatly on the location, which determines the technical factors (such as solar resources available), as well as the nontechnical (such as the country's political situation). The economics also depend on scale; at the time of writing, the cost for utility-level solar PV was less than the cost of building and operating new coal plants in the US.

Solar Thermal Applications

Solar energy thermal applications range from low-temperature applications (swimming pool heating or domestic hot water) up to medium- or high-temperature applications (space heating, absorption cooling or steam production for electrical generation). Figure 10-1 shows examples of typical installations.

The most common solar energy thermal application is for domestic hot water (DHW) production. Solar hot water can be used as the sole source of domestic hot water or to preheat incoming supply water. This can be as simple as having an uninsulated tank in a hot attic in a southern climate to having a batch water heater on the roof of a building. The same solar collectors can be used to deliver thermal energy for space heating. A typical installation for the combined production of DHW and space heating (i.e., solar combi systems) includes the solar collectors, the heat storage tank, and a boiler used as an auxiliary heater. Combi systems require a larger collector area than a DHW system to meet the higher loads. It is possible to use a heat storage tank and a DHW storage vessel, but it may also be suitable to combine them in a single storage tank (with a high vertical stratification) to meet the different operating temperatures for space heating and DHW. To assess and compare performances of different designs for solar combi systems, the International Energy Agency (IEA) launched Task 26 to address issues in this area (http://task26.iea-shc.org). Standardized classification and evaluation processes and design tools

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